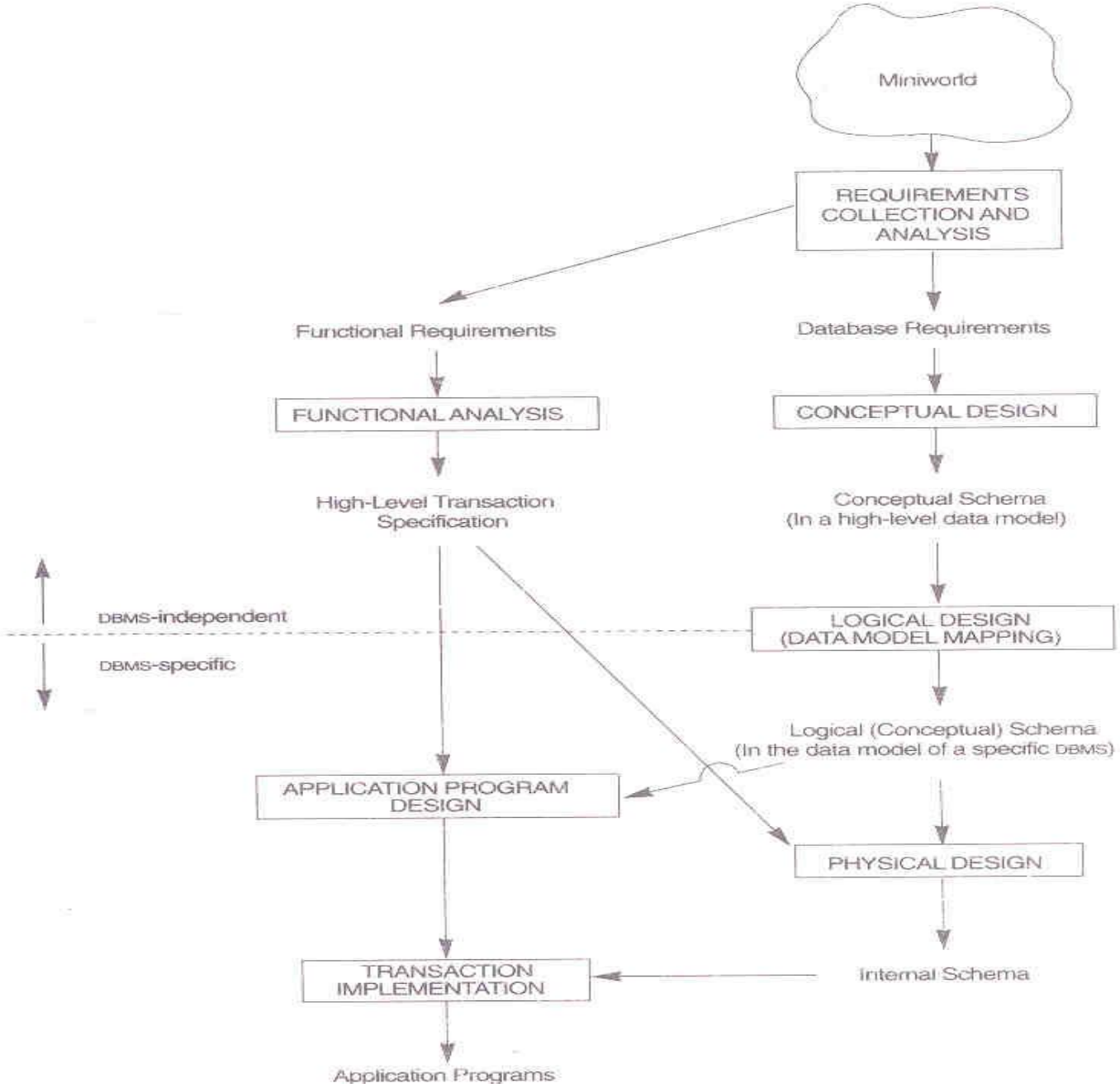
Database design phases


Miniworld
REQUIREMENTS COLLECTION AND ANALYSIS
Functional Requirements
Database Requirements
FUNCTIONAL ANALYSIS
CONCEPTUAL DESIGN
High-Level Transaction Specification
Conceptual Schema (In a high-level data model)
DBMS-independent
DBMS-specific
LOGICAL DESIGN (DATA MODEL MAPPING)
Logical (Conceptual) Schema (In the data model of a specific DBMS)
APPLICATION PROGRAM DESIGN
PHYSICAL DESIGN
TRANSACTION IMPLEMENTATION
Internal Schema
Application Programs

Database Design

The database design can be divided into six steps, and ER modeling is relevant for the first three.

1. Requirements analysis involves discussion with user groups, study of current operating environment, analysis of existing system/applications which the database is expected to replace.

Database Design

2. Conceptual database design involves a high level description of the data to be stored in the DB. This step uses ER model

3. Logical database design involves choosing a DBMS to implement our DB, and convert the conceptual design into a database schema. The ER schema is converted into a relational schema.

Database Design

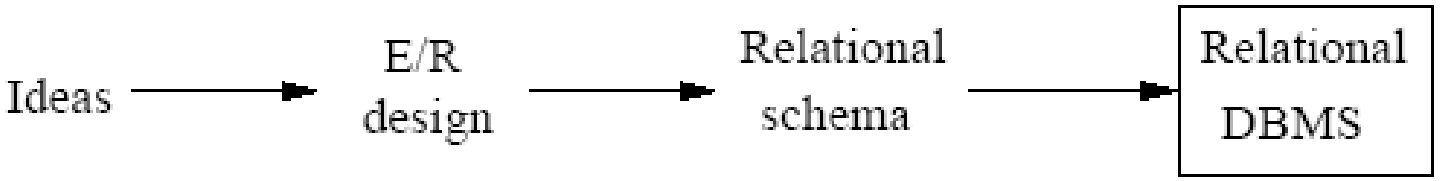
4. Schema refinement involves analysis of the DB schema for potential problems and refine it.

5. Physical DB design considers the expected workloads and refines the design so that it meets the required performance criteria. This involves building indexes, clustering some tables or substantial redesign of the schema.

Database Design

6. Security design involving identifying different user groups, the roles played by them, and the necessary parts of the DB they need to access. DBMS provides mechanisms to ensure the access is restricted accordingly to different users/groups.

Database modeling process



ER Model

- ER model has three main concepts:
  - Entities (and their entity types and entity sets)
  - Attributes (simple, composite, multivalued)
  - Relationships (and their relationship types and relationship sets)

E-R Model

- Entity
  - a "thing" (object, event) in the real world
  - physical vs. conceptual existence: car vs. job
  - *Entity* = “thing”, an abstract object of some sort.
- *Entity set* = a collection of similar entities.
  - Similar to a class in object-oriented languages.
- Examples:

  students, movies

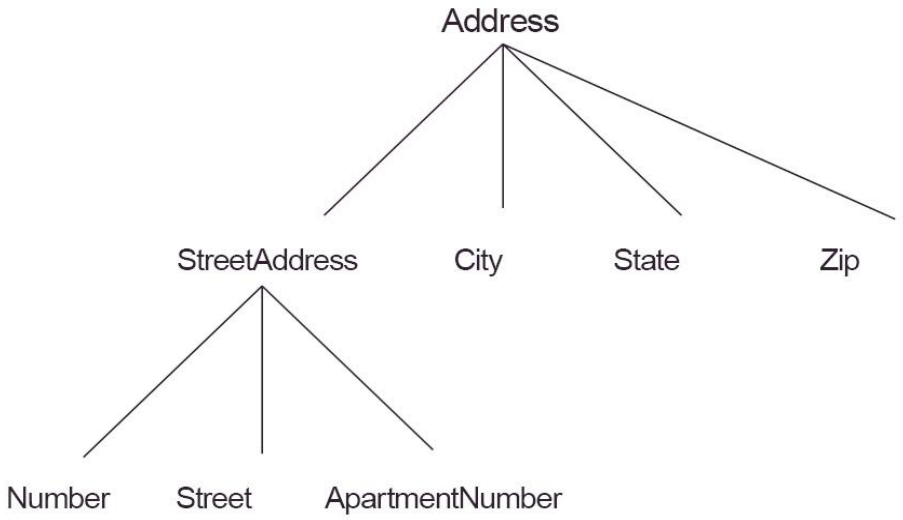
Attribute

- Attributes = properties of the entities in an entity set.
  - Generally, all entities in a set have the same properties.
  - Attributes are simple values, e.g. integers or character strings.
- Examples:

  studentNo, accountNo

Types of Attributes

- Simple vs. Composite Attributes
  - single atomic attribute vs. several component attributes (subparts) e.g. age vs. address(street, city, state, zip)
  - a hierarchy of composite attributes:



Types of Attributes

- Single-valued vs. Multi-valued Attributes
  - single value vs. multiple values for an entity
  - {CarColors}, {CollegeDegrees}
- Stored vs. Derived Attributes
  - birth date vs. age (calculated from birth date)
- Complex Attributes
  - composite and multi-valued attributes may be nested to any number of levels
  - A complex attribute: AddressPhone

{AddressPhone((Phone(AreaCode, PhoneNumber)},
Address(StreetAddress(Number, Street, ApartmentNumber), city, State, Zip))}

Entity Types

- Entity Type
  - Defines a collection of entities that have the same attributes e.g. COMPANY, EMPLOYEE
  - An entity is an instance of an entity type
  - All entities having the same set of properties are grouped into an entity type

Entity Sets & Value Sets

- Entity Set
  - The collection of all entities of a particular entity type at any point in time
- Value Set
  - Domain of values
  - A set of values allowed for an attribute
  - Each attribute can have a value from a given domain e.g. age (17-70), grade(A,B,C, D, E)

Entity Type CAR with two keys and a corresponding Entity Set

(a)

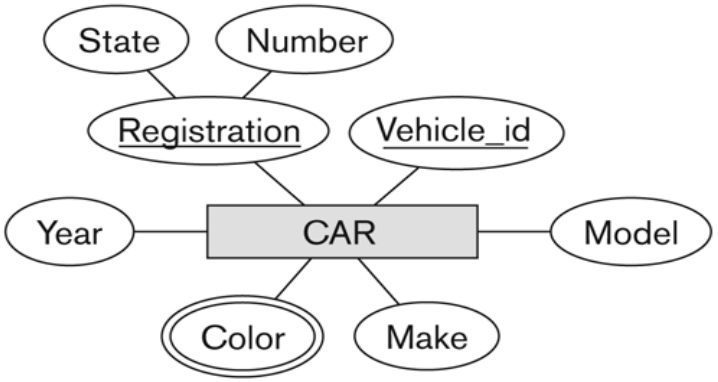


Figure 3.7
The CAR entity type with two key attributes, Registration and Vehicle_id. (a) ER diagram notation. (b) Entity set with three entities.

(b)

CAR
Registration (Number, State), Vehicle_id, Make, Model, Year, {Color}

CAR_1
((ABC 123, TEXAS), TK629, Ford Mustang, convertible, 2004 {red, black})

CAR_2
((ABC 123, NEW YORK), WP9872, Nissan Maxima, 4-door, 2005, {blue})

CAR_3
((VSY 720, TEXAS), TD729, Chrysler LeBaron, 4-door, 2002, {white, blue})

⋮

Key Attributes

- Key Attribute (Key)
  - Attribute(s) of an entity type for which each entity must have a unique value
  - Key constraint: every entity type must have at least one key
  - Simple key: VehicleID of CAR entity type
  - Composite key: Registration(State, RegistrationNumber)

Initial E-R Design

- Preliminary Conceptual Design
  - Defining the Entity Types & Attributes
  - Relationships are not included
  - Company Database:

1. An entity type DEPARTMENT with attributes Name, Number, Locations, Manager, and ManagerStartDate. Locations is the only multivalued attribute. We can specify that both Name and Number are (separate) key attributes, because each was specified to be unique

Initial E-R Design

2. An entity type PROJECT with attributes Name, Number, Location , ControllingDepartment. Both Name and Number are (separate) key attributes.
3. An entity type EMPLOYEE with attributes Name, SSN, Sex, Address, Salary, BirthDate, Department, and Supervisor. Both Name and Address may be composite attributes.
4. An entity type DEPENDENT with attributes Employee, DependentName, Sex, BirthDate, and Relationship (to the employee).

E-R Diagram Notation

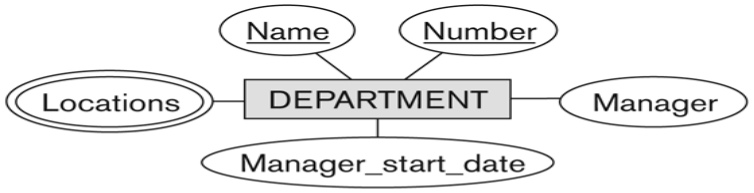
- Entity Types
  - (Strong) Entity types in rectangular boxes

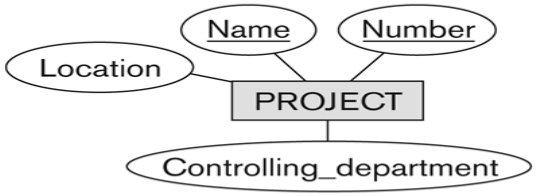


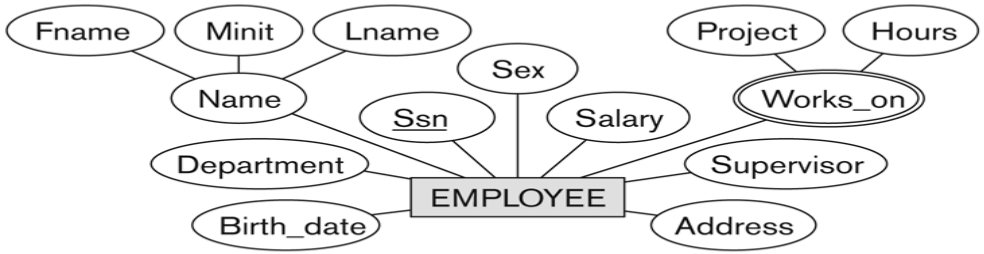
  - Weak entity types in double rectangular



Initial Design of Entity Types:
EMPLOYEE, DEPARTMENT, PROJECT, DEPENDENT







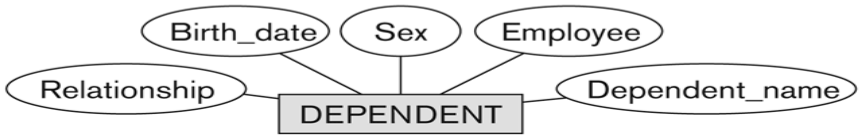


Figure 3.8
Preliminary design of entity types for the COMPANY database. Some of the shown attributes will be refined into relationships.

20

Relationship

- Relationship
  - Relates two or more distinct entities with a specific meaning
- Relationship Instance
  - An association among at least two entities belonging to one or more entity types
  - ‘John works in Accounting Department’

Relationship Types

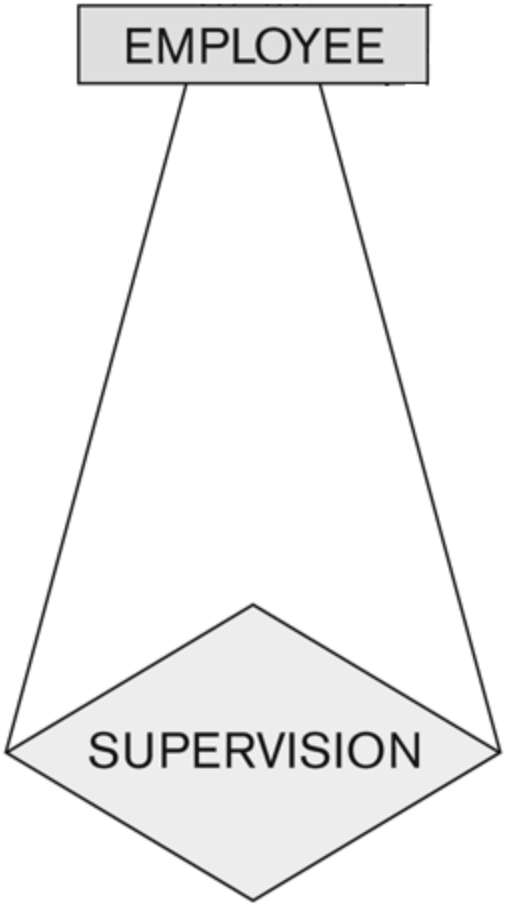
- Multiple Relationships
  - More than one relationship type can exist with the same participating entity types
  - A participating entity type plays different roles in the different relationship types
  - MANAGES (EMPLOYEE:DEPARTMENT)
  - WORKS_IN (EMPLOYEE:DEPARTMENT)

Relationship Types

- Recursive Relationship Type
  - A relationship can relate two entities of the same entity type
  - e.g.,) SUPERVISION (EMPLOYEE:EMPLOYEE)
  - SUPERVISION relationship type relates one EMPLOYEE (in the role of supervisee) to another EMPLOYEE (in the role of supervisor)

Recursive Relationship

- In some cases, the same entity type participates more than once in a relationship type in **different** roles
- Example Employee and supervised



Constrains on Relationship types

- Sometimes if we want to describe “each employee must work for exactly one department”, then we would like to describe this constrain in the schema

Relationship Constraints

- Constraints on Relationships
  - Determined from mini-world situation
    - each employee must work for only one dept.
    - an employee can work on many projects
    - every department should have one manager
    - a project is controlled by only one dept.
- Structural constraints of a relationship type
  - Cardinality Constraints
  - Participation Constraints

Cardinality of a Relationship

- The **cardinality ratio** for a binary relationship specifies the max number of relationship instances that an entity can participate in.

Cardinality of a Relationship

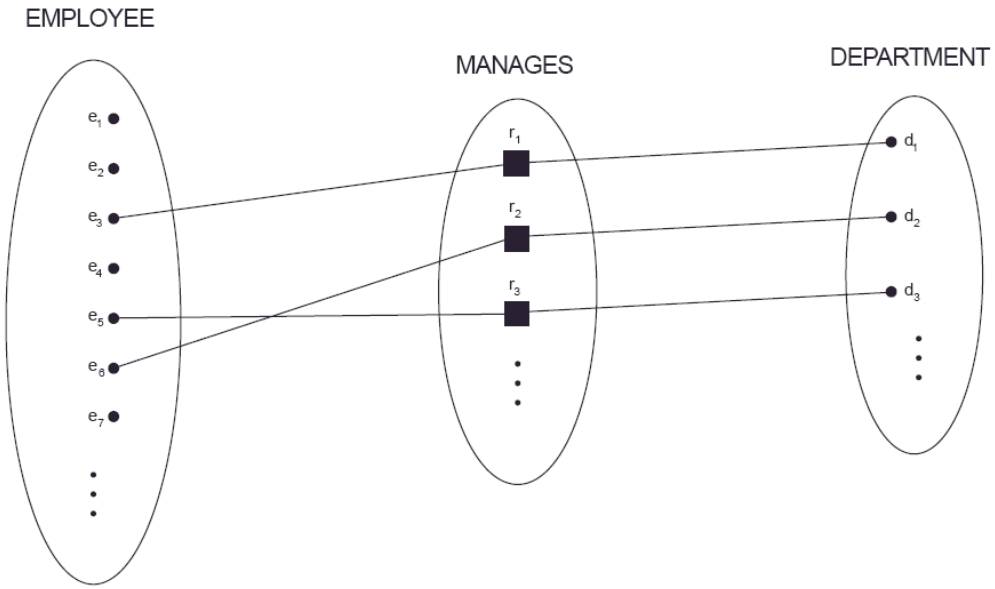
- Cardinality Ratio
  - Specify the number of relationship instances that an entity can participate in
  - 1:1 (each entity can only participate once)
  - 1:N (one entity can participate once, the other can participate many times)
  - M:N (each entity can participate many times)
  - WORKS_ON (EMPLOYEE:PROJECT) = M:N
  - WORKS_FOR (EMPLOYEE:DEPT) = N:1

Participation Constraints

- Participation Constraints
  - Specify whether each entity must participate in a relationship instance in order to exist
- Total Participation
  - Existence dependency
  - Every entity in an entity type should participate in a relationship type
    - DEPARTMENT with respect to MANAGES
    - DEPENDENT to DEPENDENTS_OF

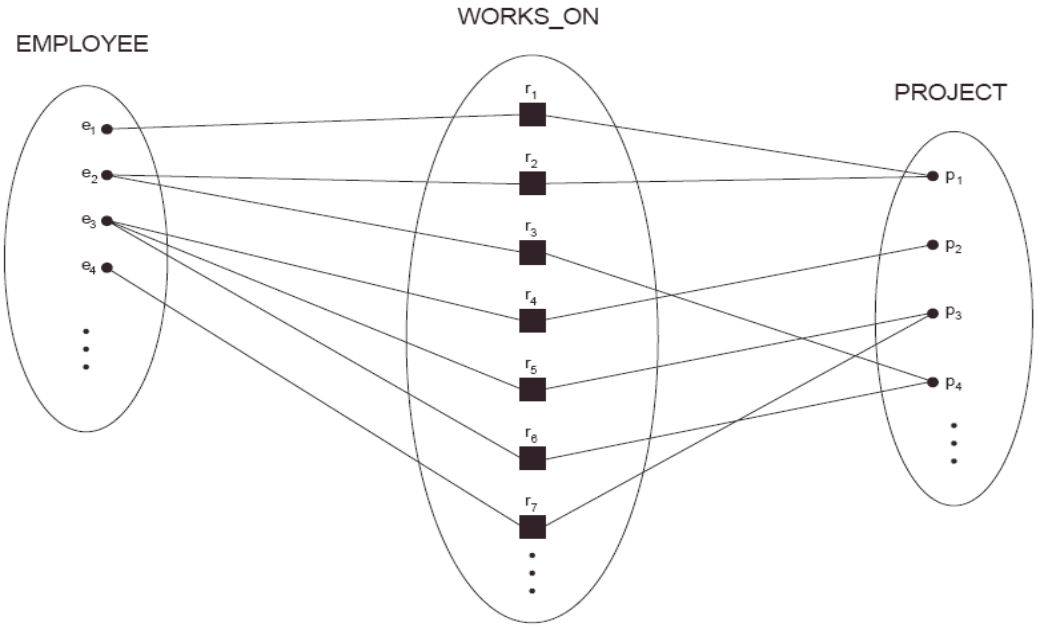
Participation Constraints

- Partial Participation
  - some entities may not participate in a relationship type
    - EMPLOYEE to MANAGES, EMPLOYEE to DEPENDENTS_OF



Attributes of Relationship Types

- Relationship types may have attributes
  - NumberOfHours for WORKS_ON
  - StartDate for MANAGES



Weak Entity Types

- Weak Entity Type
  - An entity type without a unique identifier of its own (violating key constraint)
  - Must participate in an identifying relationship type with an identifying (owner) entity type
  - Has a partial key, a set of attributes that can uniquely identify weak entities within the same owner entity
  - Has a total participation with the relationship

Weak Entity Types

- DEPENDENT vs. EMPLOYEE
  - a DEPENDENT entity is identified by the dependent's first name, and the specific EMPLOYEE that the dependent is related to
  - Weak entity type: DEPENDENT
  - Identifying (owner) entity type: EMPLOYEE
  - Identifying relationship type: DEPENDENT_OF

E-R Design Refined

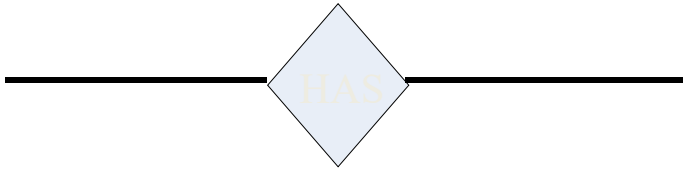
- Defining Relationship Types
  - Change the attributes that represent relationships into relationship types: Manager from DEPT., Supervisor from EMPLOYEE
  - Determine cardinality ratio and participation constraints
- Company Database Example
  - MANAGES (EMPOYEE:DEPARTMENT) = 1:1 (partial:total), has ‘StartDate’ attribute

E-R Design Refined

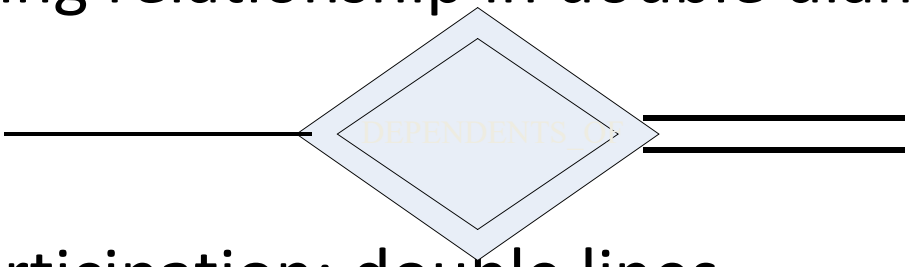
- Company Database Example
  - WORKS_FOR (EMPLOYEE:DEPT) = N:1 (t:t)
  - CONTROLS (DEPT:PROJECT) = 1:N (p:t)
  - SUPERVISION (EMPLOYEE_supervisor: EMPLOYEE_supervisee) = 1:N (p:p)
  - WORKS_ON(EMPOYEE:PROJECT) = M:N (t:t), has ‘Hours’ attribute
  - DEPENDENTS_OF (EMPOYEE:DEPENDENT) = 1:N (p:t), DEPENDENT as a weak entity type

E-R Diagram Notation

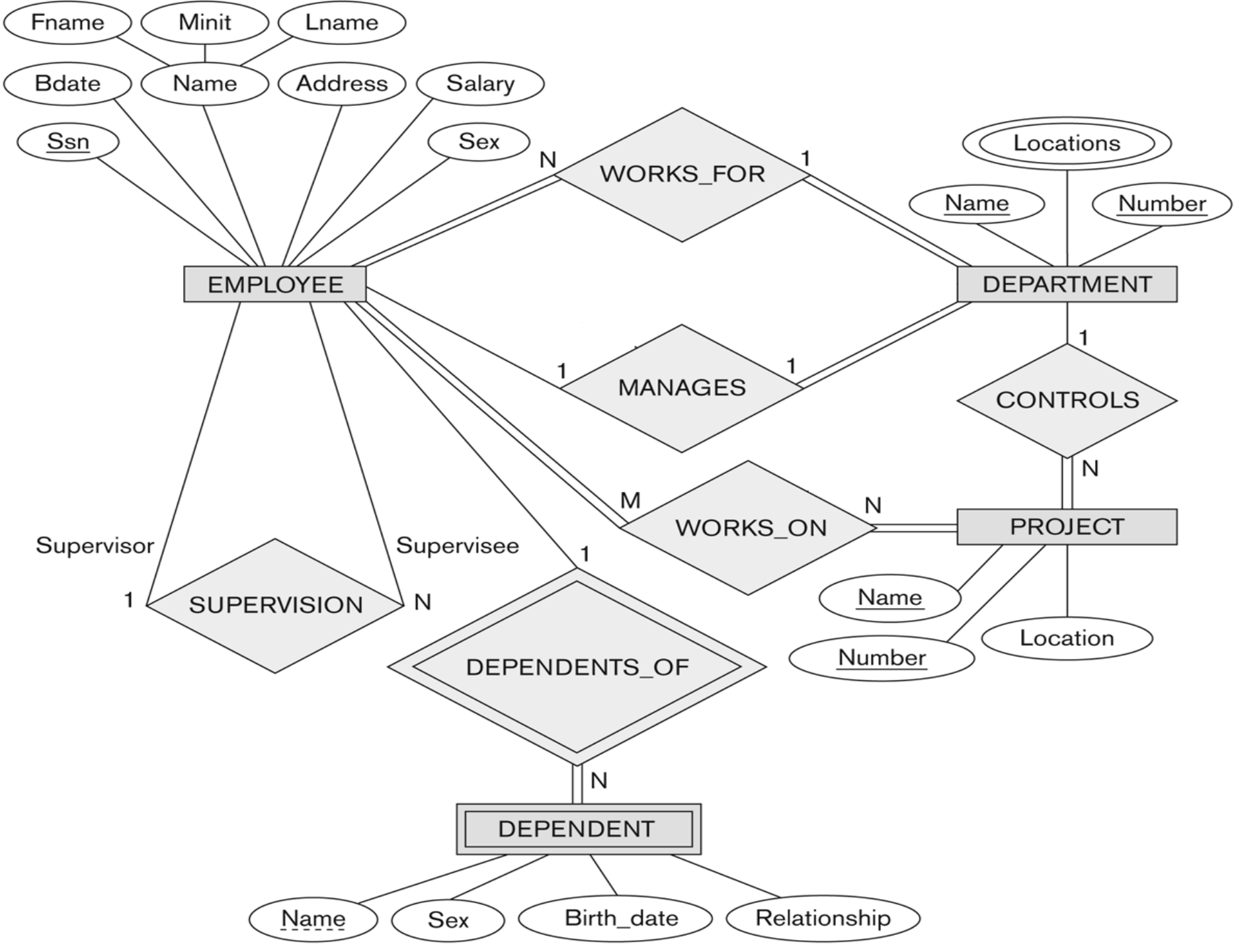
- Relationship Types
  - Relationship types in diamond-shaped boxes



  - Identifying relationship in double diamonds



  - Total participation: double lines
  - Partial participation: single line


Fname
Minit
Lname
Name
Bdate
Address
Salary
Ssn
Sex
N
WORKS_FOR
1
Locations
Name
Number
EMPLOYEE
DEPARTMENT
1
1
MANAGES
1
CONTROLS
N
M
WORKS_ON
N
PROJECT
Supervisor
Supervisee
1
1
SUPERVISION
N
Name
Location
Number
DEPENDENTS_OF
N
DEPENDENT
Name
Sex
Birth_date
Relationship